

Errol Pascua

Embedded Software Engineer — Cryptography & Security

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SUMMARY

Embedded firmware engineer with 4+ years developing bare-metal software on ARM Cortex-M microcontrollers. Core strength in low-level C, hardware-accelerated cryptographic driver development, and compliance-driven development (MISRA-C, ASPICE). Experienced building internal test frameworks, build infrastructure, and developer tooling in Python and C.

EXPERIENCE

Microchip Technology Inc.

Chandler, AZ

Software Engineer, Cryptography & Security

Jan 2022 – Present

DICE Attestation (TCG) — in progress

- Validated the platform's boot ROM DICE implementation (Layer 0) through hands-on testing: CDI generation, secure memory access, and characterizing what the hardware provides
- Drafting Layer 1 architecture and requirements for a general market solution, drawing on an existing internal DICE reference implementation on a related PIC32 device family
- Authoring ASPICE SWE.1–SWE.4 work products in parallel with the design

Hardware Cryptographic Driver Port — PIC32CM SG

- Ported hardware cryptographic drivers from 16-bit dsPIC to 32-bit ARM Cortex-M (PIC32CM SG) — same underlying IP, but required remapping hardware registers, migrating the build from XC-DSC to xc32, and resolving 32-bit architecture differences
- Drivers covered AES (ECB, CTR, GCM, CCM, XTS), SHA-1/SHA-2, CMAC, HMAC, GMAC, ECDH, ECDSA, RSA, ChaCha20, TRNG, and PKE; PKE required adapting to device-stored microcode rather than the source-embedded approach used on the 16-bit target
- Backfilled ASPICE SWE.1–SWE.3 and API reference documentation in parallel with driver work
- Modified existing secure boot use case (Python and C) to change cryptographic key storage from ROM-backed to RAM-based approach for provisioning flexibility
- Updated existing demo applications to work with latest compiled static library releases (.a files), maintaining header compatibility and verifying functionality across library version changes

Unit Testing & Verification

- Built Python-based build infrastructure to compile and run the existing Unity/CMock/CTest framework against the new device target
- Wrote all unit test suites for crypto engine, DMA, and ISR paths using hardware mocks
- Extended existing GCOV tooling with a visual HTML coverage report highlighting missing line, branch, and call coverage
- Authored ASPICE SWE.4 (unit test specifications and verification matrix)
- Debugged an NVIC interrupt collision where shared interrupt handlers caused incorrect ISR routing — identified using a logic analyzer and GPIO pin toggling, resolved by separating the handler registrations

TrustZone Secure Messaging Demo

- Developed cross-TrustZone crypto pipeline on PIC32CM SG: ECDH key agreement, TRNG IV generation, AES-256-GCM authenticated encryption, SHA-384 hashing, and ECDSA sign/verify
- Implemented NSC veneer functions and struct interfaces for secure/non-secure boundary crossing

Crypto v4 Documentation, Testing & Applications

- Authored Crypto v4 API reference documentation published to Microchip’s online developer documentation
- Ported ARTEMIS regression test framework to PIC32CX MTG, PIC32CZ CA91, PIC32CM SG, and PIC32CK SG
- Built and maintained Crypto v4 demo applications for PIC32CM SG, PIC32CX MTG, PIC32CZ CA9x, and PIC32CK SG demonstrating hardware-accelerated and software-backed cryptographic operations

Build Systems & Tooling

- Built Python/Tkinter GUI tools for MISRA-C static analysis workflows (Cppcheck Premium, Coverity) and firmware provisioning tooling
- Created a Python GUI wrapper for an existing internal firmware metadata tool (FWMDT), simplifying the interface for HSM provisioning address configuration
- Built Python CLI tooling for static analysis pipelines: configuration management (YAML/JSON parsing), XML report processing, and wrapper scripts for tool execution

Static Analysis & MISRA-C Compliance

- Progressed team toolchain from MISRA-C:2012 through MISRA-C:2023 to MISRA-C:2025
- Actively resolving 100–300+ findings per analysis cycle using Cppcheck Premium and Coverity

Collaboration & Process

- Participated in PR-based code review workflows on Bitbucket, Agile standups, and sprint ceremonies
- Coordinated cross-team dependencies with the DS PICs team on shared-IP driver integration

Southwest Engineering Concepts

Engineering Intern

Chandler, AZ

Sep 2021 – Jan 2022

- Developed embedded products from requirements through prototyping: wireless pop-time measuring instrument, vinyl film monitoring system, inductive chronograph
- Designed schematics and PCB layouts; sourced components; assembled and validated prototypes
- Programmed embedded systems in C for STM32 and dsPIC microcontrollers

EDUCATION

Arizona State University — B.S.E. Computer Systems Engineering (Cybersecurity Concentration)

May 2021

TECHNICAL SKILLS

Languages	C (primary), Python (scripting/tooling), C++ (working knowledge), Bash/Shell
Cryptography	AES (ECB/CTR/GCM/CCM/XTS), SHA-1/SHA-2, CMAC, HMAC, ECDH, ECDSA, RSA, ChaCha20, TRNG, PKE, X.509, DICE/TCG attestation
Security	TrustZone (ARMv8-M), wolfCrypt, Microchip Crypto v4, hardware-rooted trust, secure boot
Hardware	ARM Cortex-M (PIC32CM, PIC32CK, PIC32CX, PIC32CZ, SAM), dsPIC, STM32; register-level programming, ISR/DMA handling
Debug/Tools	Segger J-Link, Saleae logic analyzer, oscilloscopes, GPIO tracing, MPLAB X IDE, Segger Embedded Studio

Build Systems	CMake, xc32, GCC/MinGW, Python build automation
Static Analysis	MISRA-C:2012/2023/2025, Cppcheck Premium, Coverity
Testing	NIST ACVP, Unity, CMock, CTest, GCOV, ARTEMIS regression framework
DevOps/Collab	Git, Bitbucket, Jira, Confluence, Polarion, Tridion Docs
Process	ASPICE (SWE.1–SWE.4), Agile/Scrum